



Project 6:

CDS Pricing

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Credit Risk

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1. Compute the market-implied term structure of default rates.

We start by estimating values for the survival probabilities for the three given timeframes (0 to 1, 1 to 3, 3 to 5). We do this by using the continuous T Period extension of the formula for determining the price of a defaultable zero-coupon bond.

$$q(0, T) = \frac{1 - q(0, T) + q(0, T) \cdot R}{e^{r \cdot (T-t)}} = \frac{1}{e^{(r+s) \cdot (T-t)}}$$

With r being the risk-free rate (3.5%) and s being the spread (0.7%, 0.95%, 1.2% respectively).

First period:

After plugging in, we arrive at 0.98837 for $PS(0,1)$ (PS = survival probability) which is the survival probability for the period 0 to 1. We transform this into the default intensity by plugging into the equation:

$$PS(t, T) = e^{-\lambda(t) \cdot (T-t)}$$

Solving for $\lambda(t)$ we arrive at a default intensity of 1.17%.

Second Period:

We follow the same procedure as for the first period and we arrive at 0.95317 for the $PS(0,3)$. Following this, using the bootstrapping method, we solve for $e^{-\lambda \cdot (T-t)}$ by plugging into the following equation:

$$PS(0,3) = PS(0,1) \cdot e^{-\lambda \cdot (3-1)}$$

Solving this for λ results in 1.81%.

Third Period:

We follow the same procedure as for the first and second periods and we arrive at 0.90294 for the $PS(0,5)$.

Following this, using the bootstrapping method, we solve for $e^{-\lambda \cdot (T-t)}$ by plugging into the equation:

$$PS(0,5) = PS(0,1) \cdot e^{-0.0181 \cdot (3-1)} \cdot e^{-\lambda \cdot (5-3)}$$

from the section on the second period. Solving this for λ results in 2.71%

The final results for λ for the different time periods t :

$$\lambda(t) = \begin{cases} 0.0117 & 0 < t \leq 1 \\ 0.0181 & 1 < t \leq 3 \\ 0.0271 & 3 < t \leq 5 \end{cases}$$

2. Compute the model-implied current price (i.e., spread) of a 4-year CDS.

To price a 4-year CDS we use the survival probabilities implied by the term structure derived in part 1. With quarterly payment dates $t_i = i \times 0.25$ ($i = 1, \dots, 16$) and mid-period points $t_{mid,i} = t_i - 0.125$, the premium leg and the protection leg are:

$$PV_{prem} = \sum_{i=1}^n \left(\Delta \cdot e^{-rt_i} \cdot Q_i + \frac{\Delta}{2} \cdot e^{-rt_{mid,i}} \cdot \Delta Q_i \right)$$

$$PV_{prot} = \sum_{i=1}^n \left((1 - R) \cdot e^{-rt_{mid,i}} \cdot \Delta Q_i \right)$$

Here $\Delta Q_i = Q(t_{i-1}) - Q(t_i)$ is the marginal default probability in period i , and $n = 16$ for a 4-year CDS with quarterly coupons. The accrued-premium term (second summand in PV_{prem}) accounts for the coupon

fraction owed when default occurs between payment dates, approximated at the midpoint of each period.

The par spread is the rate s^* that sets the two legs equal:

$$s^* = \frac{PV_{prot}}{RPV01_{4yr}}$$

$$RPV01_{4yr} = \sum_{i=1}^{16} \left(\Delta \cdot e^{-rt_i} \cdot Q_i + \frac{\Delta}{2} \cdot e^{-rt_{mid,i}} \cdot \Delta Q_i \right)$$

Summing over all 16 periods:

$$RPV01_{4yr} = 3.604141$$

$$PV_{prot} = 0.040071$$

$$s^* = \frac{0.040071}{3.604141} = 0.011118 \approx 111.18 \text{ bp}$$

The model-implied par spread of a 4-year CDS on XYZ is $s^* \approx 111.18$ basis points. This lies between the 3 year (95 bp) and 5-year (120 bp) market quotes, consistent with an upward-sloping term structure of credit risk.

3. Compute the Mark-to-Market value of our current position.

The Mark-to-Market value of the existing CDS position is computed by recognizing that the protection buyer locked in a premium of 60 bp one year ago, while the current model-implied par spread for a 4-year CDS stands at 111.18 bp. Since spreads have increased, the position has gained value for the protection buyer. The *MtM* formula is:

$$MtM = (s_t - s_{old}) \times RPV01_t \times V$$

Substituting the relevant values, $s_t = 0.011118$, $s_{old} = 0.0060$, $RPV01 = 3.604141$ and $V = 5,000,000$, one obtains:

$$MtM = (0.0111 - 0.0060) \times 3.6041 \times 5,000,000 = \text{€}92,230$$

The positive sign confirms that the position yields a gain to the protection buyer, as the current market spread exceeds the contractual spread originally agreed upon.